

Data sheet

Seated valves (PN 16)

VF 2 - 2-way valve, flange

VF 3 - 3-way valve, flange

Description



VF 2 and VF 3 valves provide a quality, cost effective solution for most water and chilled applications.

The valves are designed to be combined with following actuators:

- DN 15-50 with AMV(E) 335, AMV(E) 435 or AMV(E) 438 SU actuators.
With AMV(E) 25 (SU/SD) or AMV(E) 35 actuators (with adapter **065Z0311**)
- DN 65, 80 with AMV(E) 335 or AMV(E) 435 actuators.
With AMV(E) 56 actuator (with adapter **065Z0312**)
- DN 100 with AMV(E) 55/56 or AMV(E) 65x actuators
- DN 125, 150 with AMV(E) 55/56, AMV(E) 65x, AME 685 or AMV(E) 85/86 actuators
- DN 200-300 with AME 685 or AME 855 actuators

Combinations of actuators is evident under section "Dimension".

Features:

- Soft sealing design DN15-80, 200-300
- Snap mechanical connection together with AMV(E) 335, AMV(E) 435
- Dedicated 2 and 3-port valve
- Suitable for diverting applications (3-port)

Main data:

- DN 15-300
- k_{vs} 0.63 -1350 m³/h
- PN 16
- Up to close A-AB
- Down to close A-AB (VF 3 DN 200-300)
- Temperature:
 - Circulation water/glycolic water up to 50 %:
 - 2 (-10*) ... 130 °C (DN 15-100)
 - 2 (-10*) ... 200 °C (DN 125, 150)
 - 2 (-10*) ... 130 °C (DN 200-300)
 - * At temperatures from -10 °C up to +2 °C use stem heater*
- Flange connections PN 16
- Compliance with Pressure Equipment Directive 97/23/EC

Ordering

Example:
2-way valve; DN 15; k_{vs} 1.6; PN 16;
 T_{max} 130 °C; flange connection;

- 1× VF 2 DN 15 valve
Code No.: **065Z0273**

2-way valve VF 2

DN	k_{vs} (m³/h)	T_{max} (°C)	Code No.
15	0.63	130	065Z0271
	1.0		065Z0272
	1.6		065Z0273
	2.5		065Z0274
	4.0		065Z0275
20	6.3		065Z0276
25	10		065Z0277
32	16		065Z0278
40	25		065Z0279
50	40		065Z0280
65	63		065Z0281
80	100		065Z0282
100	145	200	065B3205
125	220		065B3230
150	320		065B3255

3-way valve VF 3

DN	k_{vs} (m³/h)	T_{max} (°C)	Code No.
15	0.63	130	065Z0251
	1.0		065Z0252
	1.6		065Z0253
	2.5		065Z0254
	4.0		065Z0255
20	6.3		065Z0256
25	10		065Z0257
32	16		065Z0258
40	25		065Z0259
50	40		065Z0260
65	63		065Z0261
80	100		065Z0262
100	145	200	065B1685
125	220		065B3125
150	320		065B3150
200	630	130	065B4200
250	1000		065B4250
300	1350		065B4300

Accessories - Adapter

DN	Actuators	max.Δp (bar)	Code No.
15-50	AMV(E) 25, 35	4.0	065Z0311
65-80	AMV(E) 56	2.5	065Z0312

Accessories - Stem heater

DN	Actuators	Power supply (V/VA)	Code No.	
			Stem Heater	Adapter
15-80	AMV(E) 335, 435	24/40	065Z0315	/
15-50	AMV(E) 438 SU			enclosed
15-50	AMV(E) 25/35			065Z0311
65-80	AMV(E) 56			065Z0312
100	AMV(E) 55, 56, 65x	24/15	065Z7020	/
125, 150	AMV(E) 55, 56, 65x	24/40	065Z7022	/
125, 150	AMV(E) 85, 86, AME 685	24/20	065Z7021	/
200-300	AME 685, 855			/

Service kits

Type	DN	Code No.
Stuffing box	15	065Z0321
	20	065Z0322
	25	065Z0323
	32	065Z0324
	40, 50	065Z0325
	65, 80	065Z0327
	100	065B1360
	125, 150	065B0007
	200-300	065B3530

Data sheet

Seated valves VF 2, VF 3

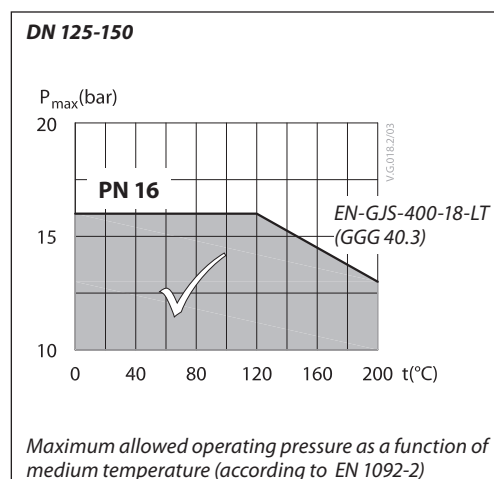
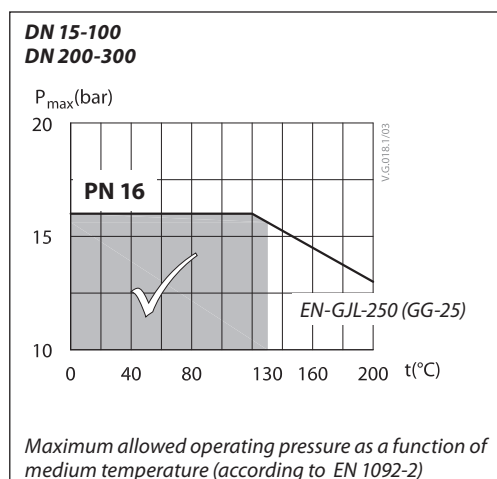
Technical data

Nominal diameter	DN	15					20	25	32	40	50	65	80	100	125	150	200	250	300			
k _{vs} value	m³/h	0.63	1.0	1.6	2.5	4.0	6.3	10	16	25	40	63	100	145	220	320	630	1000	1350			
Stroke	mm	10							15				20	30	40			57	73			
Control range		30:1	50:1				100:1													>50:1		
Control characteristic		LOG: port A-AB; LIN: port B-AB																				
Cavitation factor z		≥ 0.4															≥0.45					
Leakage	A-AB	≤ 0.03 % of k _{vs}												≤ 0.05 % of k _{vs}				≤ 0.01 % of k _{vs}				
	B-AB	≤ 1.0 % of k _{vs}																				
Nominal pressure	PN	16																				
Max. closing pressure ¹⁾		for VF 2 (up to DN 150) and for VF 3 (in mixing applications)																				
AMV(E) 335/435 (400 N)	bar	4											2.5	-								
AMV(E) 25 (SU/SD)/438 SU (450 N)																						
AMV(E) 35 (600 N)																						
AMV(E) 25 (1000 N)																						
AMV(E) 55/65x (2000 N)		-											-	1.5	1.0	0.5	-	-	-			
AMV(E) 56 (1500 N)													2.5	1.0	0.5	0.2	-	-	-			
AMV(E) 85/86 (5000 N)													-	-	3.0	1.5	-	-	-			
AME 685 (5000 N)													-	-	-	-	1.5	1.2	0.8			
AME 855 (15000N)													-	-	-	-	5.0	4.0	2.5			
Max. closing pressure ¹⁾		for VF 3 (diverting applications)																				
AMV(E) 335/435 (400 N)	bar	1											0.6	-								
AMV(E) 25 (SU/SD)/438 SU (450 N)																						
AMV(E) 35 (600 N)																						
AMV(E) 25 (1000 N)																						
AMV(E) 55/65x (2000 N)		-											-	0.3	0.6	0.5	-	-	-			
AMV(E) 56 (1500 N)													0.6	0.3	0.5	0.2	-	-	-			
AMV(E) 85/86 (5000 N)													-	-	0.6	0.6	-	-	-			
AME 685 (5000 N)													-	-	-	-	1.2	1.0	0.5			
AME 855 (15000N)													-	-	-	-	4.0	3.5	2.0			
Medium		Circulation water/glycolic water up to 50 %																				
Medium pH		Min. 7, Max. 10																				
Medium temperature ²⁾	°C	2 (-10) ... 130												2 (-10) ... 200				2 (-10) ... 130				
Connections		Flange PN 16 acc. to EN 1092-2																				
Materials																						
Valve body		Grey cast iron EN-GJL-250 (GG-25)												Ductile iron EN-GJS-400-18-LT (GGG 40.3)				Grey cast iron EN-GJL-250 (GG-25)				
Valve stem		Stainless steel																		non-magnetic stainless steel		
Valve cone		Brass										Red bronze CuSn5Zn5Pb5 (Rg 6)		GGG 40								
Stuffing box sealing		EPDM												PFTE				EPDM				

¹⁾ Maximum permissible differential pressure across the valve referred for the whole actuating range of motorised valve (a function of actuator's performance)

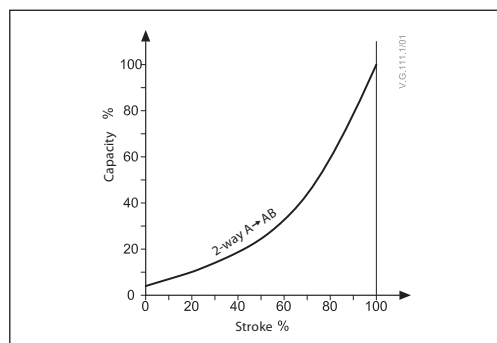
²⁾ At temperatures from -10 up to +2 °C use stem heater

Pressure temperature diagram

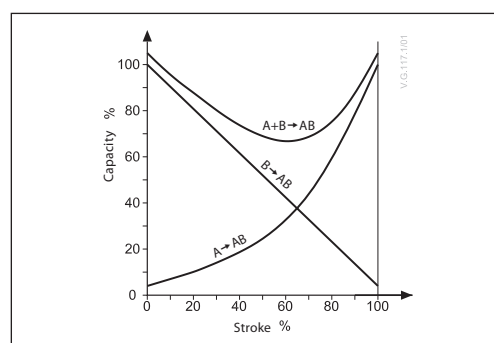


Valve characteristics

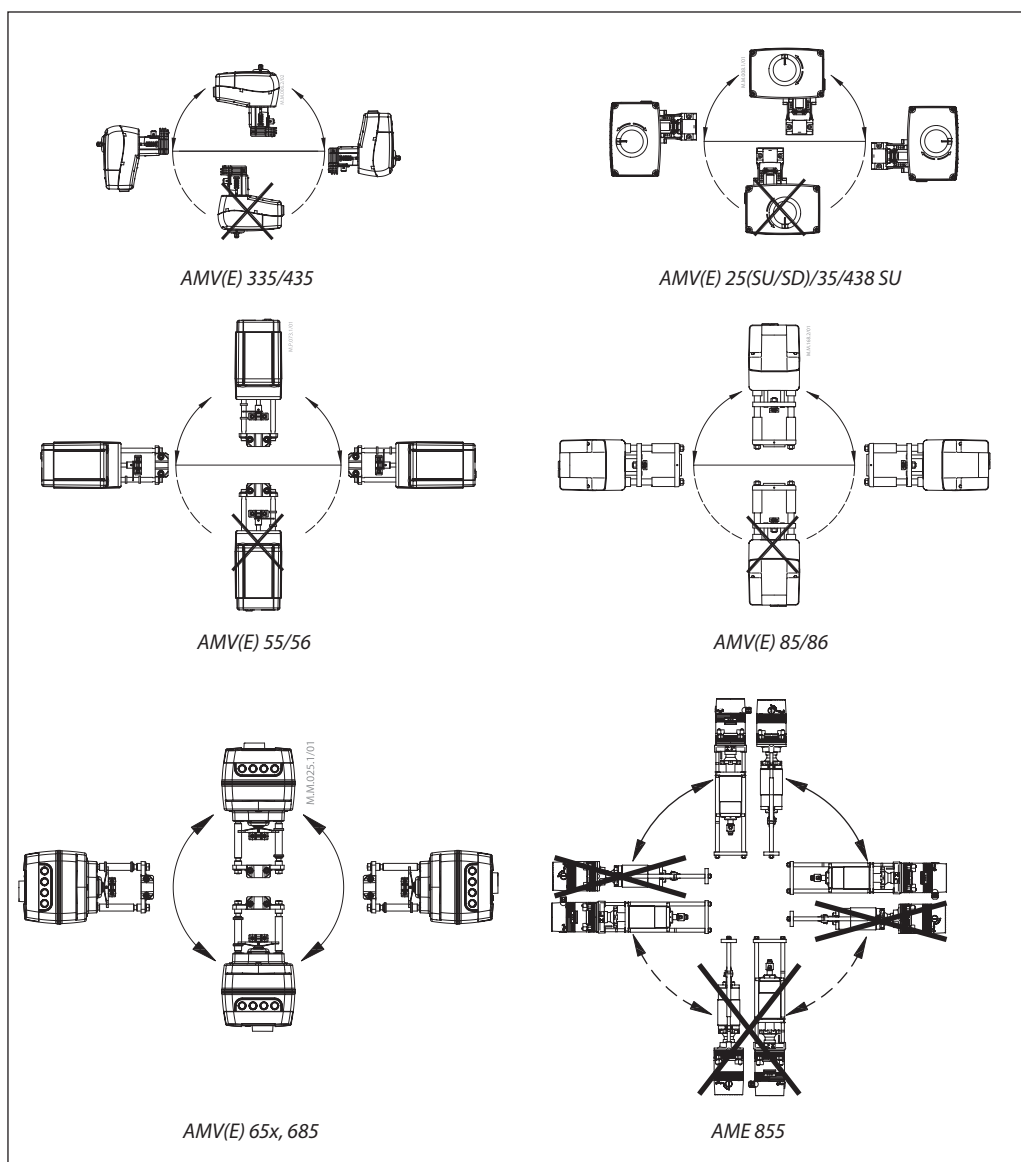
Valve characteristics log (2-way)



Valve characteristics log/lin (3-way)



Installation



Installation (continuous)

Valve mounting

Before valve mounting the pipes have to be cleaned and free from abrasion. Valve must be mounted according to flow direction as indicated on valve body.

Mechanical loads of the valve body caused by the pipes are not allowed. Valve should be free of vibrations as well.

Note:

Install a strainer upstream of the valve (e.g. Danfoss FVR/FVF)

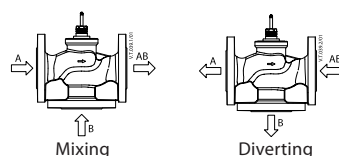
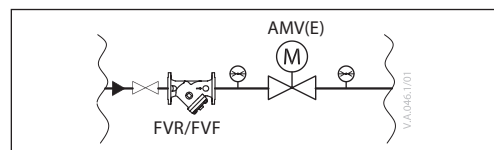


Fig. 1: Mixing or diverting connection

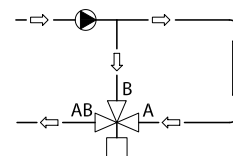


Fig. 3: Mixing valve used in diverting application

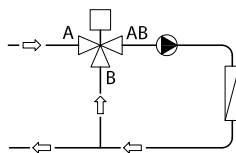


Fig. 2: Mixing valve used in mixing application

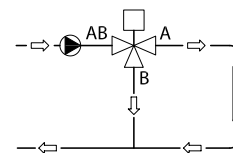


Fig. 4: Diverting valve used in diverting application

Mixing or diverting connection

3-way valve can be used either as mixing or diverting valve (fig.1).

If 3-way valve is installed as mixing valve meaning that A and B ports are inlet ports, and AB port is outlet port it can be installed in mixing (fig.2) or diverting application (fig.3).

3-way valve can be also installed as diverting valve in diverting application (fig.4) meaning that AB port is inlet and A and B ports are outlets.

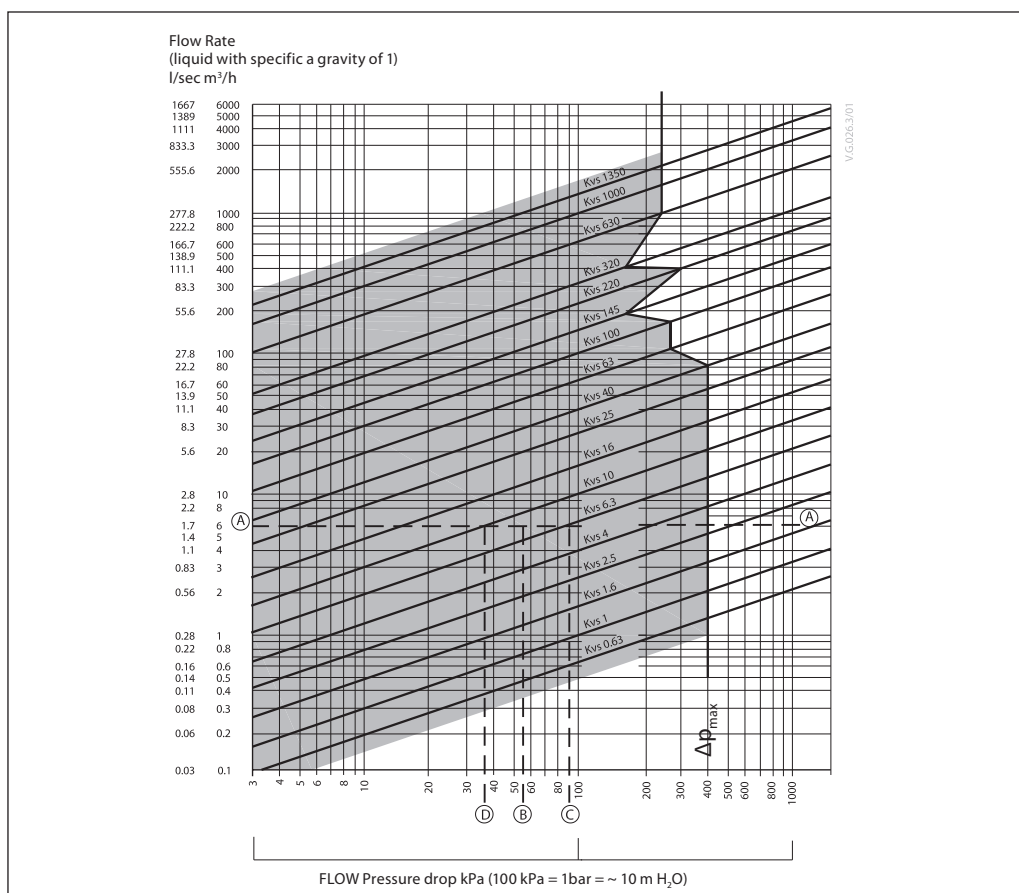
Note:

Maximal closing pressure for mixing and diverting installation are not the same. Please refer to values stated in Technical data section.

Disposal

The valve must be dismantled and the elements sorted into various material groups before disposal.

Sizing



Example

Design data:

Flow rate: 6 m³/h

System pressure drop: 55 kPa

Locate the horizontal line representing a flow rate of 6 m³/h (line A-A). The valve authority is given by the equation:

$$\text{Valve authority, } a = \frac{\Delta p_1}{\Delta p_1 + \Delta p_2}$$

Where:

Δp_1 = pressure drop across the fully open valve

Δp_2 = pressure drop across the rest of the circuit with a full open valve

The ideal valve would give a pressure drop equal to the system pressure drop (i.e. an authority of 0.5):

if: $\Delta p_1 = \Delta p_2$

$$a = \frac{\Delta p_1}{2 \times \Delta p_2} = 0.5$$

In this example an authority of 0.5 would be given by a valve having a pressure drop of 55 kPa at that flow rate (point B). The intersection of line A-A with a vertical line drawn from B lies between two diagonal lines; this means that no ideally-sized valve is available.

The intersection of line A-A with the diagonal lines gives the pressure drops stated by real, rather than ideal, valves. In this case, a valve with k_{vs} 6.3 would give a pressure drop of 90.7 kPa (point C):

$$\text{hence valve authority} = \frac{90.7}{90.7 + 55} = 0.62$$

The second largest valve, with k_{vs} 10, would give a pressure drop of 36 kPa (point D):

$$\text{hence valve authority} = \frac{36}{36 + 55} = 0.395$$

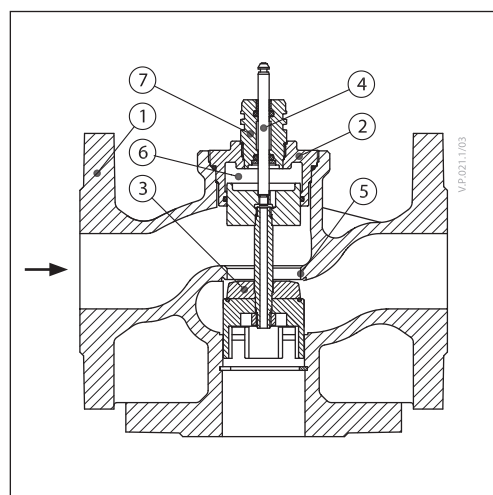
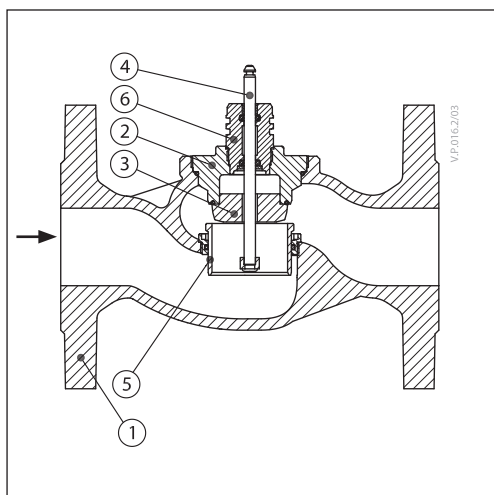
Generally, for a 3 port application, the smaller valve would be selected (resulting in a valve authority higher than 0.5 and therefore improved control). However, this will increase the total pressure and should be checked by the system designer for compatibility with available pump heads, etc. The ideal authority is 0.5 with a preferred range of between 0.4 and 0.7.

Design

(Design variations are possible)

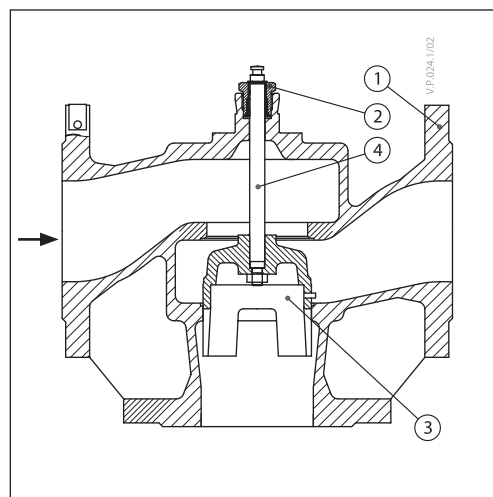
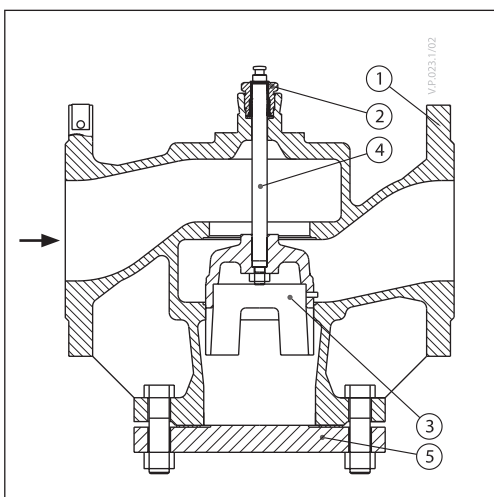
VF 2 DN 15-80

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem
5. Moving valve seat (pressure relieved)
6. Stuffing box



VF 3 DN 15-80

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem
5. Valve seat
6. Pressure relieve chamber
7. Stuffing box

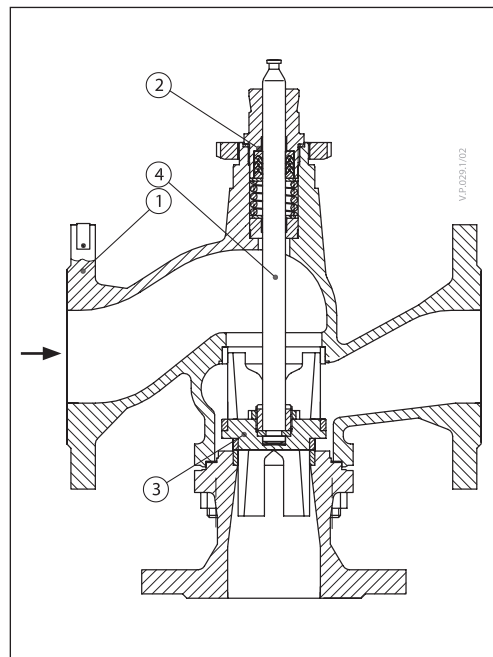
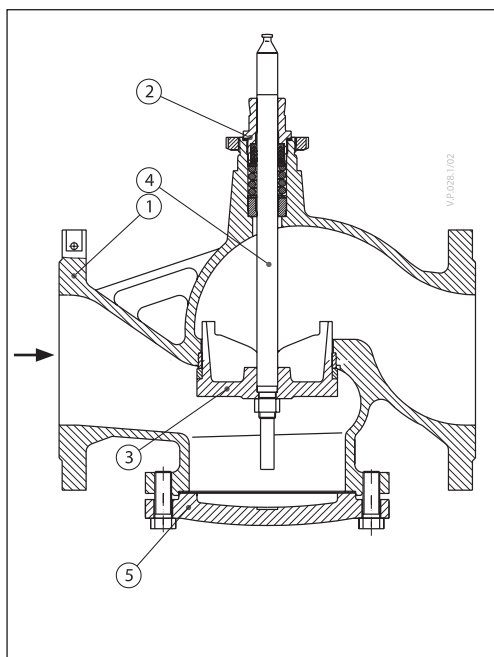


VF 2 DN 100

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem
5. Blind flange

VF 3 DN 100

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem



VF 2 DN 125-150

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem
5. Blind flange

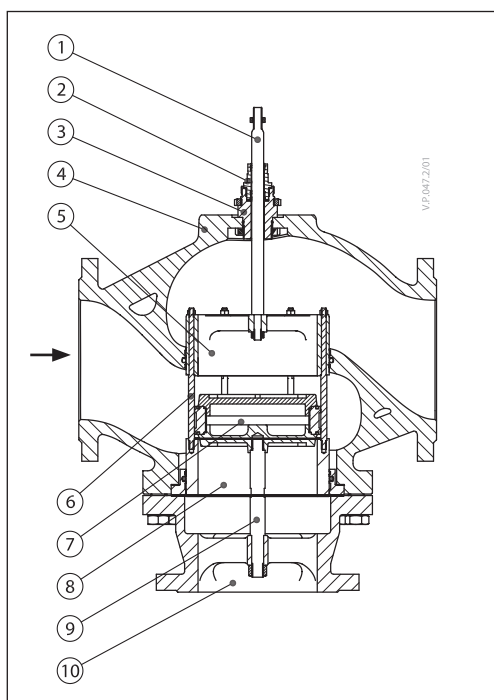
VF 3 DN 125-150

1. Valve body
2. Valve insert
3. Valve cone
4. Valve stem

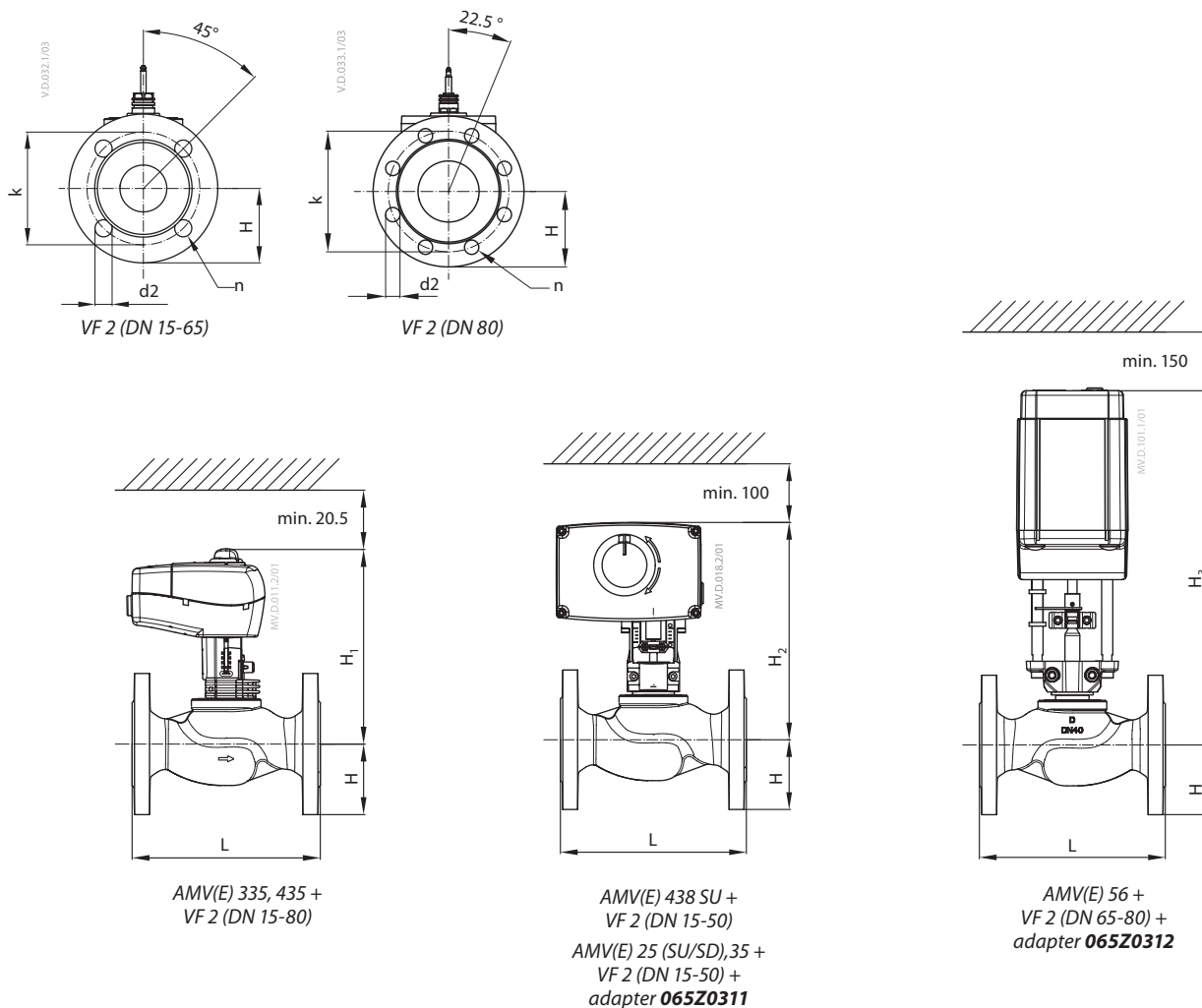
Design (*continuous*)

VF 3 DN 200-300

1. Stem
2. Stuffing box
3. Insert body
4. Valve body
5. Seat A
6. Studdle stem
7. Cone component
8. Seat B
9. Support stem
10. Valve body extension



Dimensions

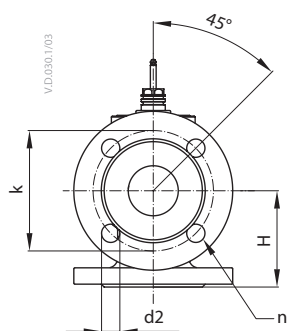


Type	DN	L	H	H ₁	H ₂	H ₃	k	d2	n	Weight (kg)
		mm								
VF 2	15	130	47.5	191	216	-	65	14	4	1.93
	20	150	52.5	194	218	-	75	14	4	2.65
	25	160	57.5	197	222	-	85	14	4	3.23
	32	180	70	202	226	-	100	19	4	4.97
	40	200	75	213	237	-	110	19	4	6.59
	50	230	82.5	218	242	-	125	19	4	8.53
	65	290	92.5	254	-	428	145	19	4	15.92
	80	310	100	258	-	432	160	19	8	18.13

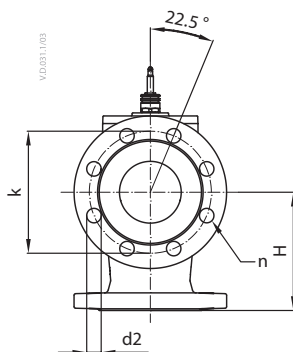
Note:

If stem heater is used dimension H₁ is increased for 28 mm and H₂ for 32 mm.

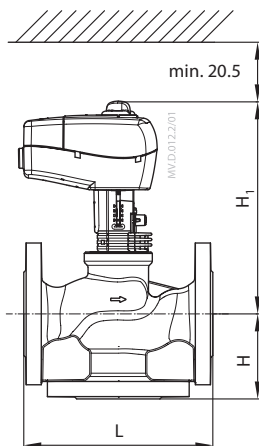
Dimensions (continued)



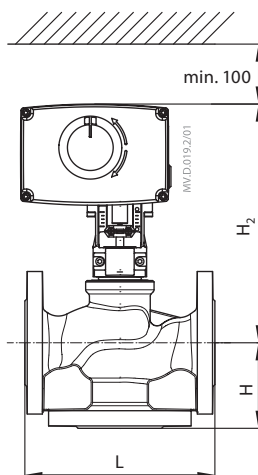
VF 3 (DN 15-65)



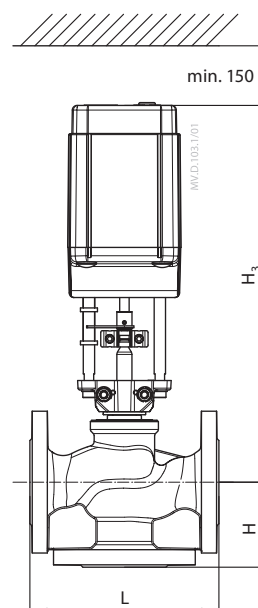
VF 3 (DN 80)



AMV(E) 335, 435 +
VF 3 (DN 15-80)



AMV(E) 438 SU +
VF 3 (DN 15-50)
AMV(E) 25 (SU/SD), 35 +
VF 3 (DN 15-50) +
adapter **065Z0311**



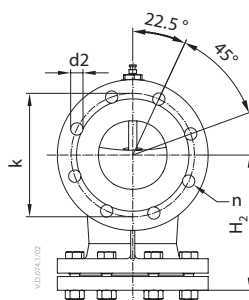
AMV(E) 56 +
VF 3 (DN 65-80) +
adapter **065Z0312**

Type	DN	L	H	H ₁	H ₂	H ₃	k	d2	n	Weight (kg)
		mm								
VF 3	15	130	63	191	216	-	65	14	4	2.61
	20	150	70	194	218	-	75	14	4	3.55
	25	160	75	197	222	-	85	14	4	4.54
	32	180	80	202	226	-	100	19	4	6.90
	40	200	90	230	255	-	110	19	4	9.05
	50	230	100	243	267	-	125	19	4	12.79
	65	290	120	254	-	428	145	19	4	19.18
	80	310	155	270	-	444	160	19	8	23.73

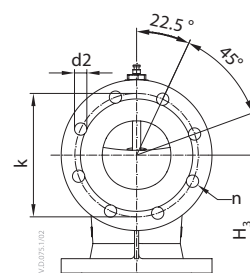
Note:

If stem heater is used dimension H₁ is increased for 28 mm and H₂ for 32 mm.

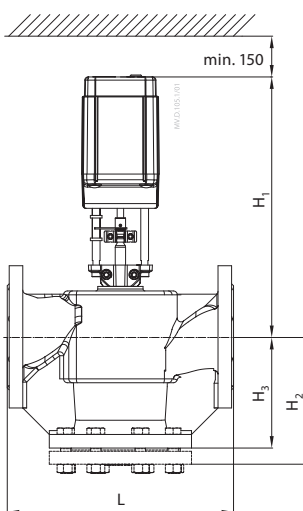
Dimensions (continued)



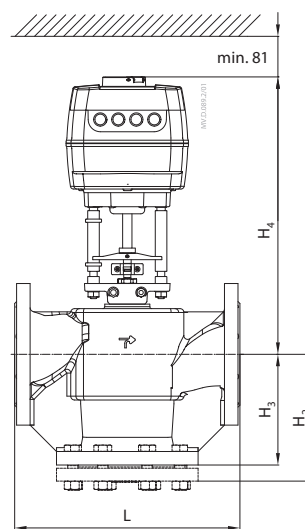
VF 2 (DN 100)



VF 3 (DN 100)



AMV(E) 55, 56 +
VF 2, VF 3 (DN 100)



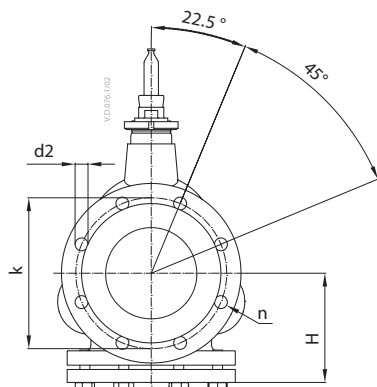
AMV(E) 65x +
VF 2, VF 3 (DN 100)

Type	DN	L	H ₁	H ₂	H ₃	H ₄	k	d2	n	Weight (kg)
		mm								
VF 2	100	350	406	196	175	450	180	18	8	39.0
VF 3										34.0

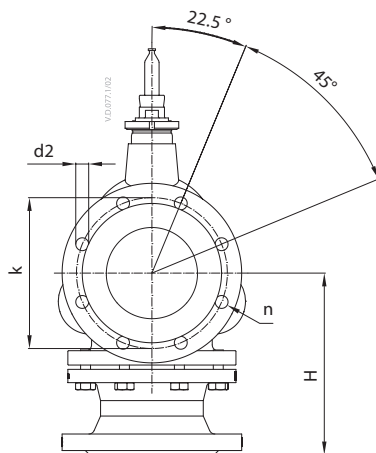
Note:

If stem heater is used dimension H remains the same.

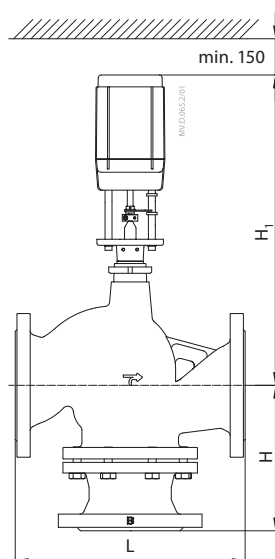
Dimensions (continued)



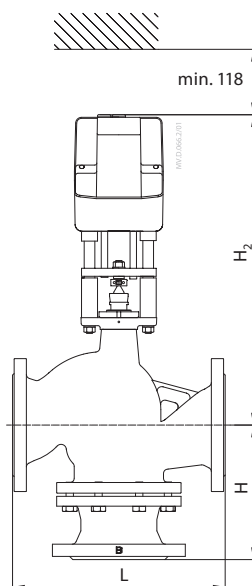
VF 2 (DN 125, 150)



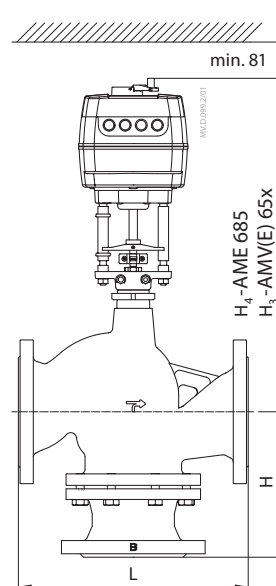
VF 3 (DN 125, 150)



AMV(E) 55, 56 +
VF 2, VF 3 (DN 125, 150)



AMV(E) 85, 86 +
VF 2, VF 3 (DN 125, 150)



AMV(E) 65x, AMV 685 +
VF 2, VF 3 (DN 125, 150)

Type	DN	L	H	H ₁	H ₂	H ₃	H ₄	k	d2	n	Weight (kg)
		mm									
VF 2	125	400	160	555	629	595	723	210	18	8	54.0
	150	480	200	560	682	648	723	240	22	8	79.0
VF 3	125	400	250	555	629	595	723	210	18	8	65.3
	150	480	300	560	682	648	723	240	22	8	92.0

Note:

If stem heater is used dimensions H₁ and H₂ remain the same.

Dimensions (continued)

